



MCP Credit Report Demo

OPENSIFT / KUBERNETES ARCHITECTURE

This document describes how the three-process MCP Credit Report Demo application is deployed on Red Hat OpenShift using Kubernetes primitives — Deployments, Services, Routes, Secrets, and ConfigMaps — within a single OpenShift Project (Namespace).

OPENSIFT 4.X

KUBERNETES

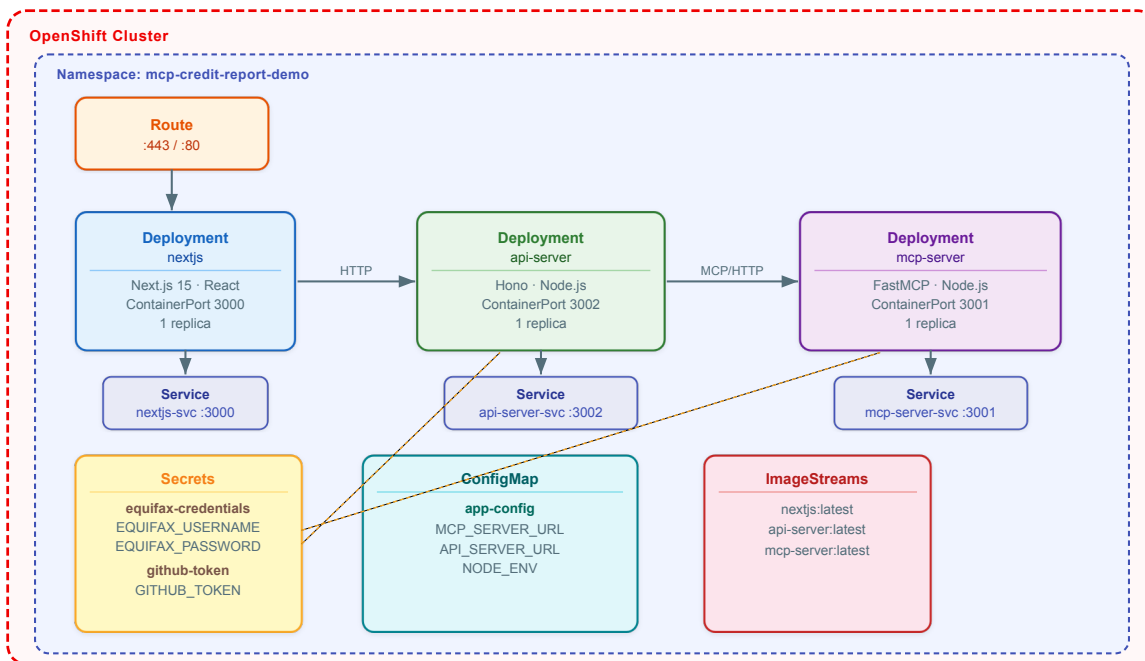
NODE.JS / TYPESCRIPT

FASTMCP · HONO · NEXT.JS



Cluster Overview

All three application processes run inside a single OpenShift Project (Namespace).

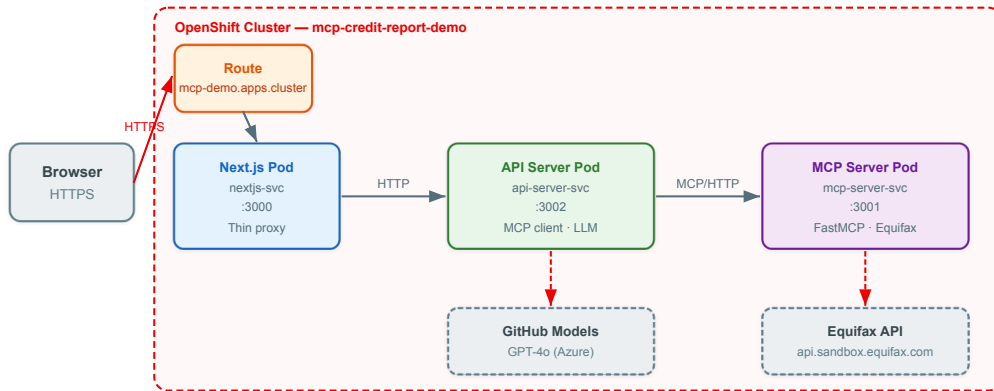


- Deployment (Pod)
- Service (ClusterIP)
- Route (Ingress)
- Secret
- ConfigMap
- ImageStream



Network Flow

Request path from the public internet through the OpenShift cluster to external APIs.



INTERNAL TRAFFIC

All pod-to-pod traffic stays within the cluster using ClusterIP Services. No external exposure except through the Route.

EXTERNAL TRAFFIC

Outbound HTTPS to Equifax sandbox and GitHub Models APIs. An `EgressNetworkPolicy` should whitelist these endpoints.

TLS TERMINATION

The Route handles TLS termination using an OpenShift-managed certificate (Let's Encrypt via cert-manager or edge termination).



Kubernetes Resources

All resources live in the `mcp-credit-report-demo` namespace.

DEPLOYMENTS

NAME	IMAGE	CONTAINER PORT	REPLICAS	ENV VARS
nextjs	nextjs:latest	3000	1	CONFIGMAP: API_SERVER_URL
api-server	api-server:latest	3002	1	CONFIGMAP: MCP_SERVER_URL SECRET: GITHUB_TOKEN
mcp-server	mcp-server:latest	3001	1	SECRET: EQUIFAX_USERNAME SECRET: EQUIFAX_PASSWORD

SERVICES (CLUSTERIP)

NAME	SELECTOR	PORT	TARGET PORT	PROTOCOL
nextjs-svc	app=nextjs	3000	3000	TCP
api-server-svc	app=api-server	3002	3002	TCP
mcp-server-svc	app=mcp-server	3001	3001	TCP

ROUTE

FIELD	VALUE
Host	mcp-demo.apps.<cluster-domain>
Service	nextjs-svc
Port	3000

SECRETS

SECRET NAME	KEYS	USED BY
equifax-credentials	EQUIFAX_USERNAME EQUIFAX_PASSWORD	mcp-server
github-token	GITHUB_TOKEN	api-server

FIELD	VALUE
TLS	Edge termination

Insecure traffic	Redirect to HTTPS
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Request Flow & Build Pipeline

Step-by-step request lifecycle and how container images reach the cluster.

REQUEST FLOW

- 1

Browser → Route
User submits the credit report form. Browser sends HTTPS POST to the OpenShift Route hostname.
- 2

Route → Next.js Service
Route terminates TLS and forwards the request to `nextjs-svc:3000` inside the cluster.
- 3

Next.js → API Server
The Next.js API route proxies the request to `api-server-svc:3002` with no business logic.
- 4

API Server → MCP Server
API server connects to `mcp-server-svc:3001/mcp` via `StreamableHTTP` and calls the `get_credit_report` tool.
- 5

MCP Server → Equifax API
Fetches OAuth token then POSTs to the Equifax OneView sandbox. Returns extracted JSON.
- 6

API Server → GitHub Models
Sends the JSON summary to GPT-4o via GitHub Models for plain-English summarisation.
- 7

Response flows back
Markdown summary returns through the chain. Browser renders styled HTML via react-markdown.

BUILD & DEPLOY PIPELINE

- 1

Source → BuildConfig
Three `BuildConfig` objects use the Node.js S2I builder image to compile TypeScript and produce container images.
- 2

ImageStream tags updated
Successful builds push to the internal OpenShift registry and update `:latest` ImageStream tags.
- 3

Deployment rollout triggered
ImageStream triggers cause Deployments to roll out new pods automatically with zero downtime.

CONFIGMAP: APP-CONFIG

KEY	VALUE	USED BY
<code>MCP_SERVER_URL</code>	<code>http://mcp-server-svc:3001</code>	<code>api-server</code>
<code>API_SERVER_URL</code>	<code>http://api-server-svc:3002</code>	<code>nextjs</code>
<code>NODE_ENV</code>	<code>production</code>	All pods



Resource Summary

Complete inventory of OpenShift / Kubernetes objects required to deploy the application.

KIND	NAME	PURPOSE	TYPE / NOTES
NAMESPACE	mcp-credit-report-demo	Isolates all application resources	OpenShift Project
DEPLOYMENT	nextjs	Next.js 15 React UI + thin proxy	1 replica · port 3000
DEPLOYMENT	api-server	Hono API server — MCP client + LLM	1 replica · port 3002
DEPLOYMENT	mcp-server	FastMCP server — Equifax data fetching	1 replica · port 3001
SERVICE	nextjs-svc	ClusterIP for Next.js pod	ClusterIP · TCP 3000
SERVICE	api-server-svc	ClusterIP for API server pod	ClusterIP · TCP 3002
SERVICE	mcp-server-svc	ClusterIP for MCP server pod	ClusterIP · TCP 3001
ROUTE	mcp-demo-route	Exposes Next.js to the internet with TLS	Edge TLS · HTTPS redirect
SECRET	equifax-credentials	Equifax OAuth client credentials	Opaque · env injection
SECRET	github-token	GitHub Models (GPT-4o) API token	Opaque · env injection
CONFIGMAP	app-config	Service URLs and runtime config	env injection
IMAGESTREAM	nextjs	Tracks Next.js container image	Triggers rollout on update
IMAGESTREAM	api-server	Tracks API server container image	Triggers rollout on update
IMAGESTREAM	mcp-server	Tracks MCP server container image	Triggers rollout on update
BUILDCONFIG	nextjs-build	S2I build for Next.js	nodejs S2I builder

KIND	NAME	PURPOSE	TYPE / NOTES
BUILDCONFIG	api-server-build	S2I build for API server	nodejs S2I builder
BUILDCONFIG	mcp-server-build	S2I build for MCP server	nodejs S2I builder
NETWORKPOLICY	allow-internal	Allows pod-to-pod traffic within namespace	Ingress from same namespace
NETWORKPOLICY	allow-route-ingress	Allows Route/HAProxy to reach Next.js	Ingress from router namespace